

MediStock – Milestone 2 Document

Medicine Inventory and Supplier Management System

1. Introduction

The MediStock Medical Inventory Management System is a full-stack application designed to simplify medicine inventory management for pharmacies and healthcare organizations. The system helps users manage medicines, suppliers, stock quantities, expiry information, and inventory search.

During Week 4, the main focus is to develop and integrate the medicine inventory APIs, supplier management system, medicine dashboard, stock tracking, and medicine search and filtering functionality.

2. Objectives

The objectives of this milestone are:

- Develop REST APIs for medicine inventory management.
- Implement complete supplier management functionality.
- Create an interactive medicine inventory dashboard.
- Track medicine stock quantities accurately.
- Implement medicine search and filtering.
- Integrate the frontend with backend APIs.
- Test the major inventory and supplier operations.

3. Task 1: Implement Medicine Inventory APIs

The first task is to develop backend APIs for managing medicines.

Functional Requirements:

- Add a new medicine.
- View all medicines.
- Retrieve a medicine using its ID.

- Update medicine details.
- Delete a medicine.
- Validate medicine information.
- Test all APIs to ensure they work correctly.

Medicine information may include:

- Medicine ID
- Medicine Name
- Category
- Manufacturer
- Batch Number
- Expiry Date
- Price
- Quantity
- Minimum Stock Level
- Supplier Name

Medicine API Operations

POST /medicines — Add medicine

GET /medicines — View all medicines

GET /medicines/{id} — Get medicine by ID

PUT /medicines/{id} — Update medicine

DELETE /medicines/{id} — Delete medicine

Expected Result:

The medicine APIs should successfully perform CRUD operations and return appropriate responses for successful operations and invalid requests.

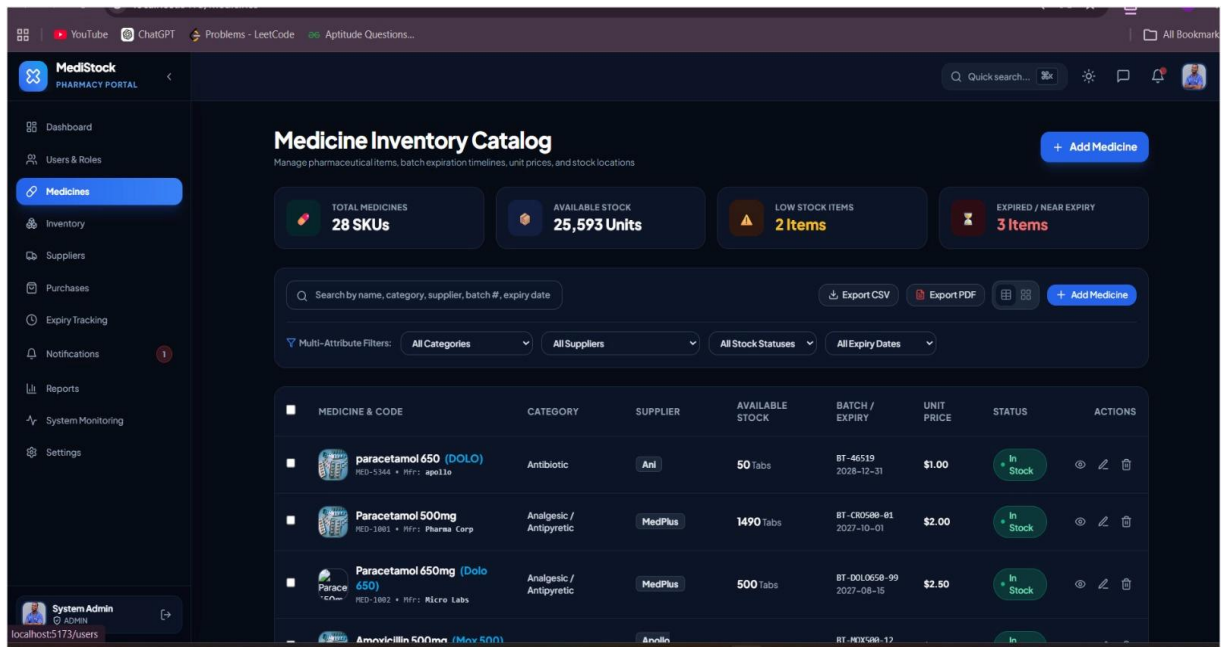


Fig 1-Medicine Inventory Catalog

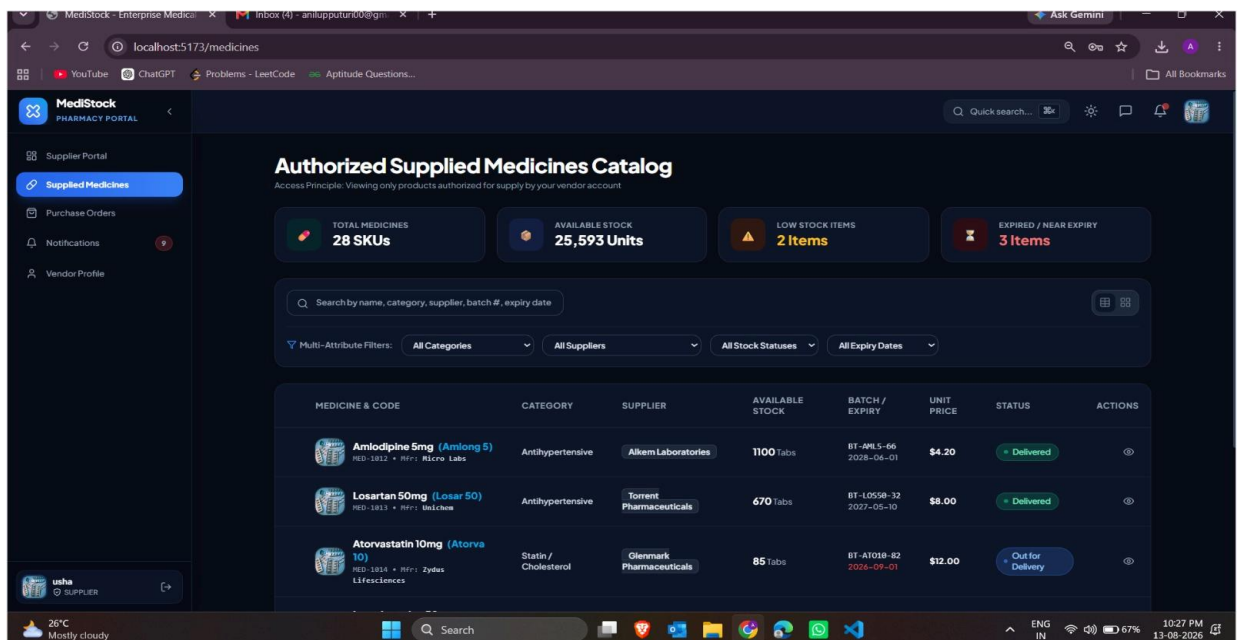


Fig 2-Authorised Supply Medicine

4. Task 2: Build the Supplier Management System

The supplier management module is responsible for maintaining supplier information and connecting suppliers with the medicine inventory.

Users should be able to:

- Add new suppliers.
- View the supplier list.
- Edit supplier details.
- Delete suppliers.
- Search suppliers.
- Associate suppliers with medicines where required.

Supplier information can include:

- Supplier ID
- Supplier Name
- Contact Person
- Phone Number
- Email
- Address

Expected Result:

The supplier management page should provide complete CRUD functionality and allow users to maintain supplier information efficiently.

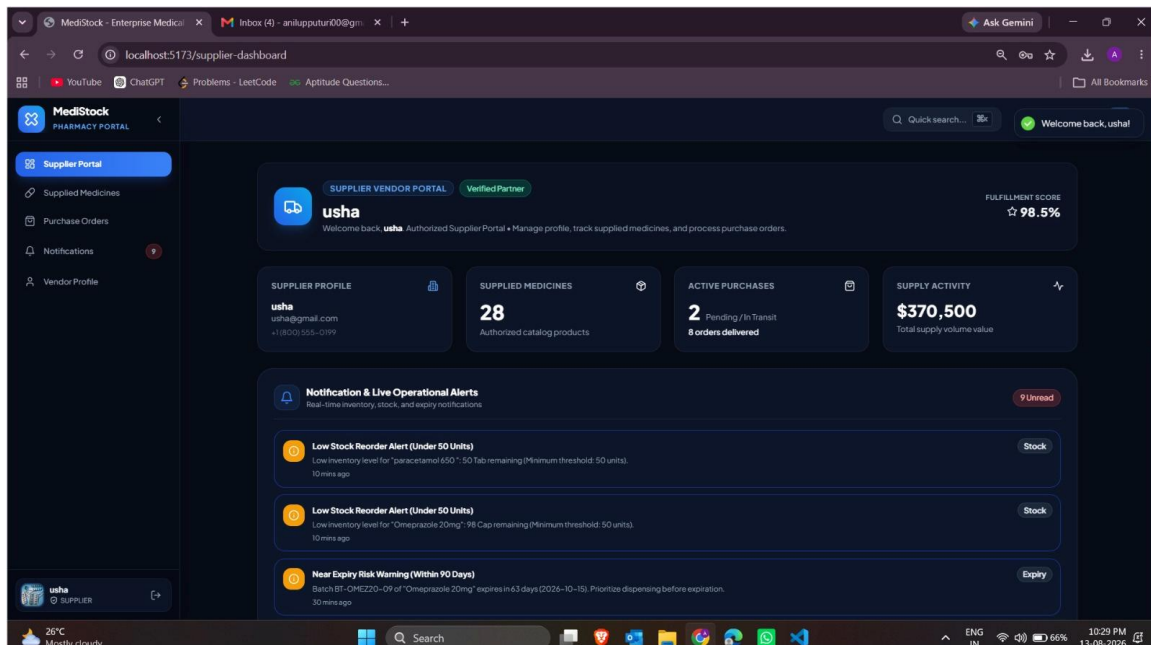


Fig 3-Supplier Dashboard

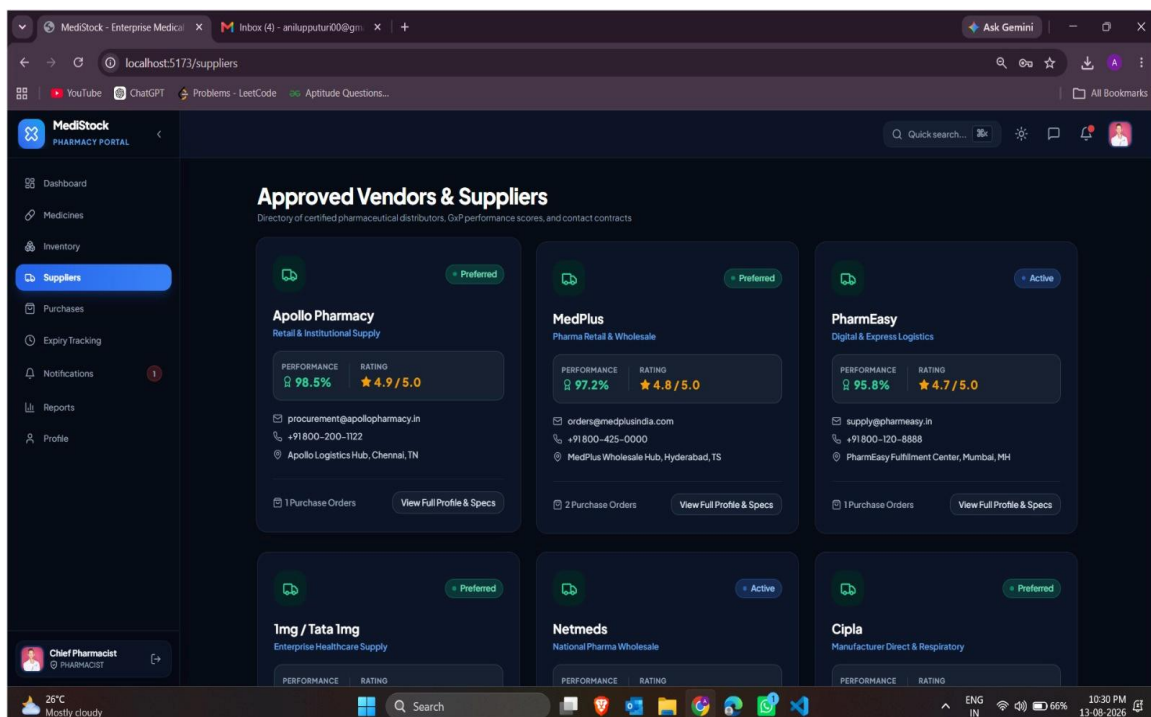


Fig 4-Approved Suppliers

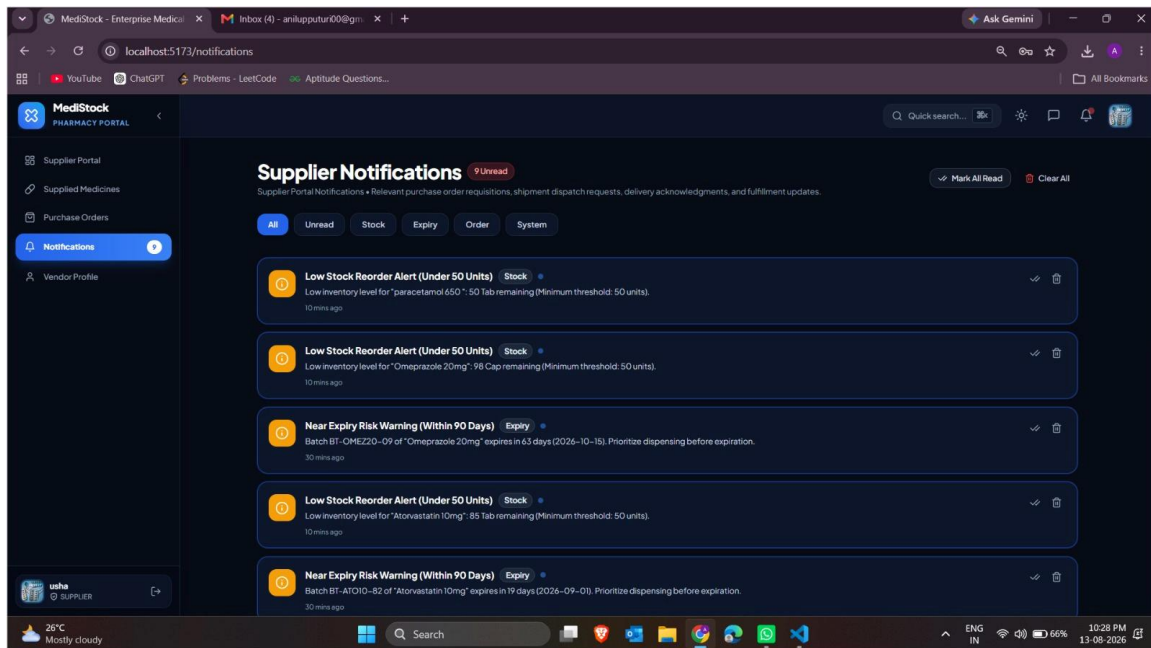


Fig 5-Supplier Notifications

5. Task 3: Create Medicine Dashboard Pages

The medicine dashboard should provide a quick overview of the current inventory.

Dashboard components:

- Total Medicines — total number of medicine records.
- Available Stock — medicines with sufficient stock.
- Low-Stock Medicines — medicines below the minimum stock level.
- Expired Medicines — medicines whose expiry date has passed.
- Near-Expiry Medicines — medicines approaching expiry, if implemented.

The dashboard should be clean, responsive, easy to navigate, integrated with backend APIs, and updated when inventory information changes.

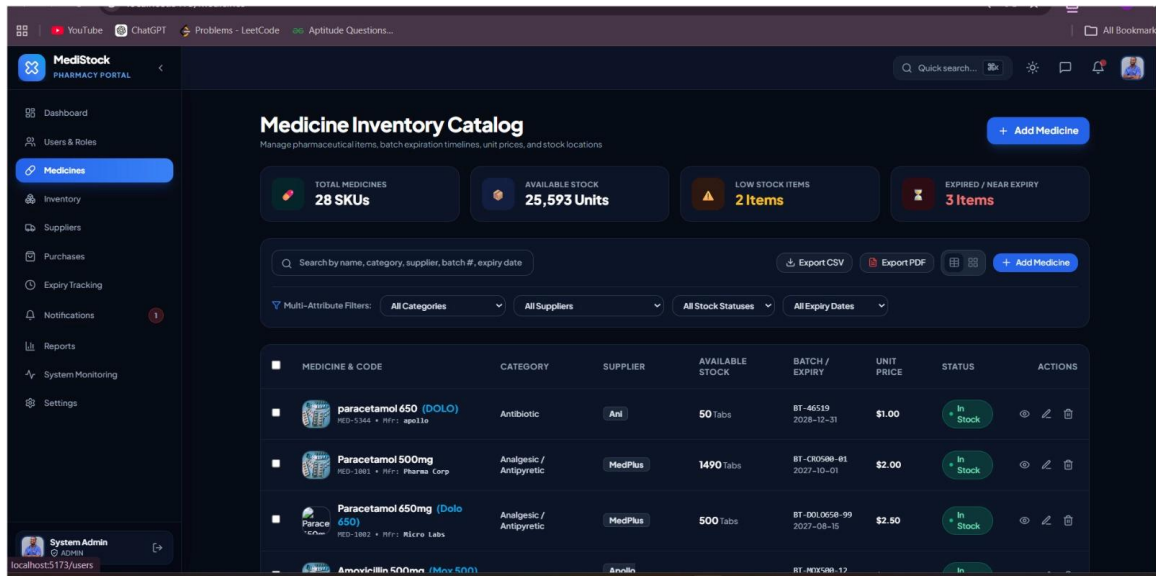


Fig 6-Medicine Dashboard

6. Task 4: Implement Stock Tracking Functionality

The stock tracking functionality ensures that the current quantity of every medicine is accurately maintained.

Stock should be updated whenever:

- A new medicine is added.
- Medicine is issued or sold.
- Medicine is restocked.
- Medicine quantity is manually updated.

Example:

Initial stock = 100

Issued = 20

Current stock = 80

After restocking 50 units:

$80 + 50 = 130$

Current stock = 130

Low Stock Detection:

The system compares the current quantity with the minimum stock level.

For example, if current quantity is 5 and minimum stock level is 10, the medicine is classified as Low Stock.

MediStock PHARMACY PORTAL

Inventory Audit & Stock Logs
Complete immutable ledger of stock additions, clinical disbursements, disposal write-offs, and adjustments

Search log by medicine, staff, or reason... Type: All Movements

LOG ID	MEDICINE ITEM	ACTION TYPE	QUANTITY CHANGE	PREVIOUS STOCK	NEW STOCK	PERFORMED BY	TIMESTAMP	CLINICAL REASON
LOG-4758	paracetamol 650	Stock Out	-50	100	50	Current User	2026-08-13 11:58:24	Routine stock replenishment
LOG-6938	paracetamol 650	Stock In	+2	98	100	Current User	2026-08-13 11:56:06	Routine stock replenishment
LOG-3499	paracetamol 650	Stock Out	-2	100	98	Current User	2026-08-13 11:55:53	Routine stock replenishment
LOG-6879	paracetamol 650	Stock In	+100	0	100	Current User	2026-08-13 11:50:26	Initial medicine entry added to database
LOG-6092	Paracetamol 500mg	Stock Out	-7	1497	1490	Current User	2026-08-12 19:58:18	Routine stock replenishment
LOG-5684	Paracetamol 500mg	Stock Out	-3	1500	1497	Current User	2026-08-12 18:35:45	Routine stock replenishment
LOG-9695	paracetamol 650	Stock Out	-10	910	900	Current User	2026-08-12 18:11:29	Routine stock replenishment

Fig 7-Stock Tracking

7. Task 5: Add Medicine Search and Filtering

Users should be able to quickly find medicines using:

Search:

- Medicine name
- Category
- Supplier

Filters:

- Available Stock
- Low Stock
- Expired Medicines
- Near-Expiry Medicines, if implemented

Expected Result:

Search and filtering should display matching medicines efficiently without requiring users to manually browse the complete inventory.

MediStock
PHARMACY PORTAL

Quick search...

Expiration & Batch Risk Control

Cold-chain and room-temperature pharmaceutical expiration timelines with automated warnings

Timeline View | Calendar View

NEAR EXPIRY WARNING (3 BATCHES)
Expiring within the next 90 days

EXPIRED WRITE-OFFS (1 BATCHES)
Quarantined for disposal

Near Expiry Medicines Countdown

Warning (63d) Batch: BT-OME220-09
Omeprazole 20mg
Gastrointestinal
Stock Count: **98 Caps**
Location: Shelf B-03 [Inspect >](#)

Critical (19d) Batch: BT-ATO10-02
Atorvastatin 10mg
Statin / Cholesterol
Stock Count: **85 Tabs**
Location: Shelf E-03 [Inspect >](#)

Critical (12d) Batch: BT-LOX500-MEXP
Levofloxacin 500mg
Antibiotic
Stock Count: **280 Tabs**
Location: Shelf B-05 [Inspect >](#)

Quarantined Expired Stock

MEDICINE ITEM	BATCH NUMBER	STOCK QTY	EXPIRY DATE	LOCATION	STATUS PILL
Dicycline 100mg	BT-DIC100-EXP	140 Caps	2026-08-01	Shelf B-04	Expired

System Admin
ADMIN
localhost:5173/notifications

26°C Mostly cloudy

Search

ENG IN 66% 10:25 PM 13-08-2026

Fig 8-Expiry Control

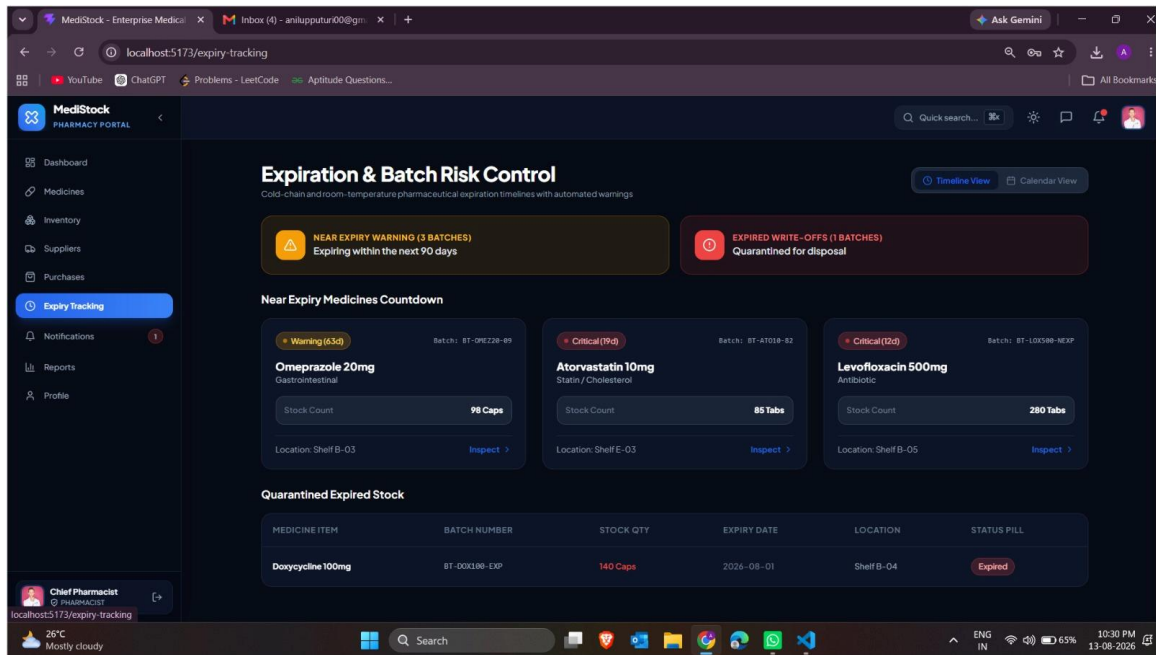


Fig 9-Expiry Tracking

8. Technology Used

Frontend:

- React.js
- JavaScript
- HTML
- CSS
- Axios
- React Router

Backend:

- Java
- Spring Boot
- Spring Data JPA
- Spring Security
- REST APIs

Database:

- MySQL

Development and Testing:

- Visual Studio Code / NetBeans
- Maven
- Postman
- Git and GitHub

9. System Integration

The frontend communicates with the Spring Boot backend through REST APIs.

Basic flow:

User → React Frontend → REST API → Spring Boot Backend → Database

Example – Add Medicine:

The user enters medicine information → React sends a POST request → Spring Boot receives the request → Medicine is saved in the database → Response is returned to the frontend → Dashboard information is updated.

10. Testing

Medicine Testing:

- Add medicine.
- View all medicines.
- View medicine by ID.
- Update medicine.
- Delete medicine.
- Test invalid medicine ID.
- Test invalid or missing medicine information.

Supplier Testing:

- Add supplier.
- View suppliers.
- Update supplier.

- Delete supplier.
- Test supplier information validation.

Stock Testing:

- Add stock.
- Reduce stock.
- Restock medicine.
- Verify low-stock detection.
- Verify dashboard stock counts.

Search and Filter Testing:

- Search by medicine name.
- Filter by category.
- Filter by supplier.
- Filter available medicines.
- Filter low-stock medicines.
- Verify expired/near-expiry medicines.

11. Expected Outcomes by the End of Milestone 2

By the end of this milestone, the MediStock system should demonstrate:

1. A functional medicine inventory management system.
2. Working medicine CRUD APIs.
3. A functional supplier management module.
4. Supplier and medicine association where required.
5. A medicine inventory dashboard.
6. Accurate stock tracking.
7. Low-stock identification.
8. Expiry and near-expiry identification where implemented.
9. Medicine search functionality.
10. Medicine filtering functionality.
11. Frontend and backend integration.
12. Successful testing of major system functions.

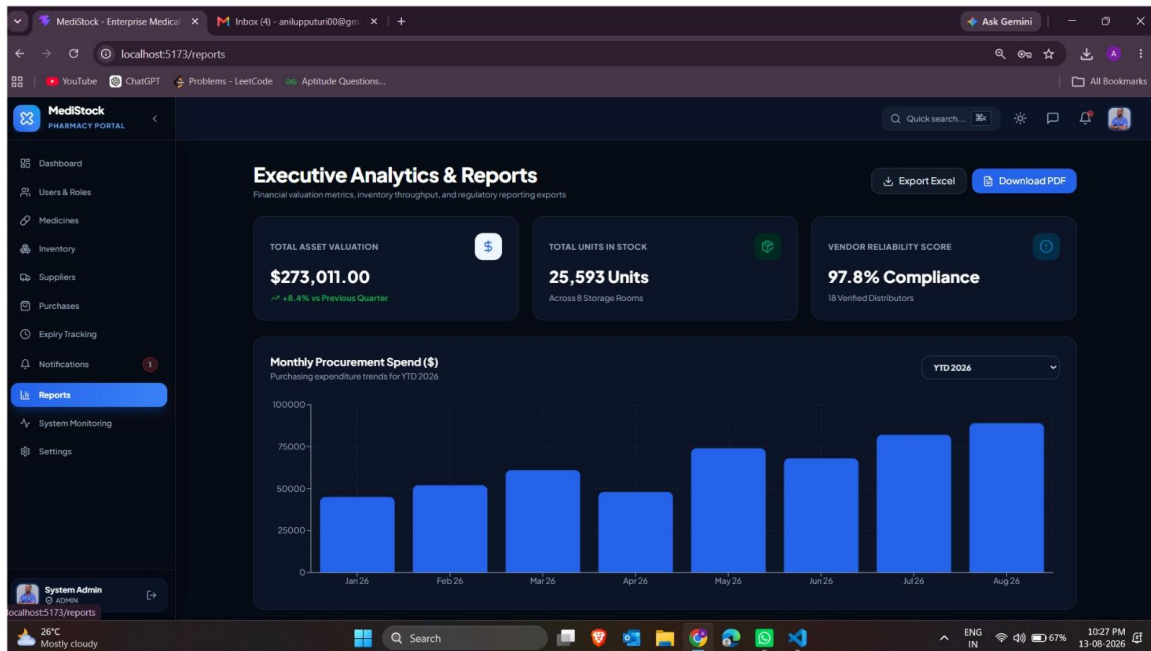


Fig 10-Reports

Admin Purchase Management & History
Purchase medicine stock, calculate total costs, assign stock to pharmacists, and manage purchase history

Export Report + Purchase Medicine

Search by POID, medicine, pharmacist, supplier

All Suppliers All Pharmacists All Statuses T1 DESC

PO ID / PHARMACIST	MEDICINE NAME	SUPPLIER	QUANTITY & TOTAL	PURCHASE DATE	BATCH / EXPIRY	STATUS	ACTIONS
PO-8985 Dr. Sarah Jenkins	Pantoprazole 40mg (Pantocid 40)	Netmeds	\$27,000.00 5000 units	Aug 11, 2026	BT-PAN40-33 Exp: Jul 28, 2027	Completed	Restocked
PO-8989 Dr. Sarah Jenkins	Calcium + Vitamin D3 (Shekal 500)	Torrent Pharmaceuticals	\$42,000.00 4000 units	Aug 11, 2026	BT-SHE500-11 Exp: Sep 1, 2028	Completed	Restocked
PO-8981 JohnDoe	Paracetamol 650mg (Dolo 650)	HedPlus	\$25,000.00 10000 units	Aug 10, 2026	BT-DOL650-99 Exp: Aug 11, 2027	Approved	
PO-8986 JohnDoe	Cetirizine 10mg	Img / Tata Img	\$14,000.00 4000 units	Aug 9, 2026	BT-CET10-88 Exp: Sep 18, 2028	Completed	Restocked
PO-8982 Dr. Sarah Jenkins	Amoxicillin 500mg (Mox 500)	Apollo Pharmacy	\$42,500.00 10000 units	Aug 8, 2026	BT-MOX500-12 Exp: Nov 18, 2027	Completed	Restocked
PO-89818 JohnDoe	Ibuprofen 400mg (Brufen 400)	Abbott	\$32,000.00 8000 units	Aug 7, 2026	BT-BRU400-55 Exp: Sep 18, 2028	Completed	Restocked
PO-8987 Dr. Sarah Jenkins	Salbutamol 4mg (Asthalin)	Cipla	\$25,000.00 10000 units	Aug 6, 2026	BT-AS14-05 Exp: Sep 18, 2027	Shipped	In-Transit
PO-89813	Azithromycin 500mg (Aithral)		\$44,000.00	Aug 6, 2026	BT-AZ1500-04	Completed	Restocked

Fig 11-Purchase History

12. Milestone Completion Criteria

The Week 4 milestone can be considered successfully completed when:

- Medicine CRUD APIs are implemented and tested.
- Supplier CRUD functionality is working.
- The inventory dashboard displays correct information.
- Stock quantities update correctly after inventory operations.
- Low-stock medicines are identified automatically.
- Medicine search works correctly.
- Medicine filtering works correctly.
- Frontend pages communicate successfully with the backend.
- The system provides accurate and usable inventory information.

13. Conclusion

The Week 4 milestone establishes the core inventory management functionality of the MediStock platform. The implementation of medicine APIs, supplier management, dashboard pages, stock tracking, and search and filtering provides the foundation for efficient medical inventory management.

Once these features are fully tested and integrated, the system will be capable of managing medicines and suppliers while providing users with an accurate and easy-to-use view of inventory status.